UDCT v7.13 | POST-REGISTRATION II

Hydrus I Jeans Sign-Boundary and Hessian Reproducibility Audit

Won Shik Paik | Independent Researcher | 22 October 2026

REPOSITORY AND CITATION LINKS

GitHub repository:

https://github.com/wspaik5/UDCT_v7.13_Postregistration_II_Reproducibility_Audit

Zenodo DOI (reserved; activated upon publication):

https://doi.org/10.5281/zenodo.23559995

ARCHIVED RESULTS (not new physical validation)

24 candidate brackets | 48 endpoint occurrences | 288 saved profiles

18 robust-positive | 30 unresolved | 0 robust-negative endpoints

0 validated strict sign-transition brackets

B01: native-grid minimum ~ -0.158964; common-grid minimum +0.231148.

These minima use different radial domains and are not directly comparable.

No new 3D modified-gravity Hessian solution or star-by-star Hydrus I fit.

REPRODUCIBILITY SCOPE

Retrospective saved-array checks and archival provenance.

The underlying physical solver is not independently rerun in this package.

The original preregistration plan is included without modification.

Manuscript and figures follow. DOI registration requires Zenodo publication.

UDCT v7.13 Post-registration II: Hydrus I Jeans Sign-Boundary and Hessian Reproducibility Audit

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Publication status: Upload-ready manuscript; DOI reserved, not yet registered by publication.
already published.

Original Post-registration I: https://doi.org/10.5281/zenodo.23539127
Original preregistration: https://doi.org/10.5281/zenodo.23342971

Abstract

We conduct a retrospective, post-hoc reproducibility audit of the archived UDCT v7.13 six-arm stationary Jeans sign-boundary tests using observation-informed Hydrus I inputs. The 24 inherited candidate brackets produce 48 endpoint occurrences and 288 saved six-arm common-radius profiles (1,200 radial samples each). The archived final outcomes remain 18 robust-positive, 30 unresolved and zero robust-negative endpoint occurrences; zero of 24 strict opposite-sign brackets is validated. An independently written checker confirms no common-radius six-arm negative witness among saved profiles and all 48 common-grid minima exceed their computed tau thresholds. This necessary-condition audit does not reconstruct the native-grid minima needed for the original final decision. The preregistration plan and eight frozen inherited inputs were reconciled by SHA-256. Repeated endpoint occurrences are not independent physical tests; their precise identity depends on the full parameter tuple. We clarify the B01 native/common radius-domain discrepancy and limitations in original source recovery. No self-consistent three-dimensional total-potential Hessian or star-by-star Hydrus I likelihood fit is demonstrated. A proposed v7.14 preregistration is outlined without representing it as an executed experiment.

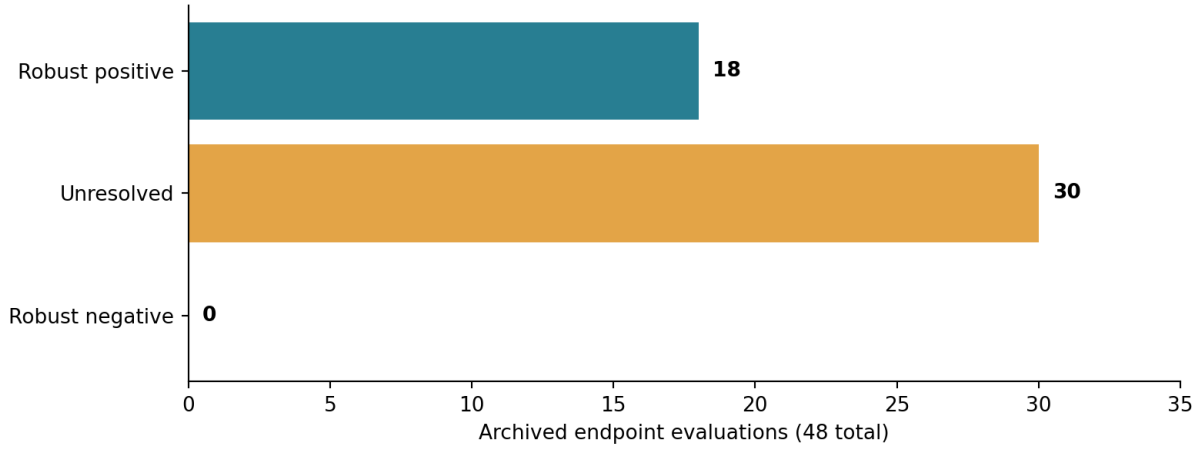


Figure 1. Archived final classifications across 48 endpoint occurrences. These are not 48 statistically independent galaxy tests.

1. Scope and scientific background

The spherical stationary anisotropic Jeans equation, d(nu sigma_r^2)/dr + 2 beta nu sigma_r^2/r = -nu g(r), is evaluated under specified screened modified-gravity prescriptions and tracer assumptions. Hydrus I supplies observation-informed model scales, not a fit to the complete observed velocities of individual stars. A negative numerical second moment within a chosen closure does not constitute a measured negative stellar velocity dispersion. Higher-derivative GQUMOND has established prior art (Milgrom 2023); this study does not claim invention of its Hessian terms.

2. Materials, original preregistration and audit boundaries

The frozen v7.12 files, v7.13 preregistration plan, original v7.13 Post-registration I Hessian archive and post-hoc Phase 3 evidence form the audit record. Eight source/table hashes match the preregistration plan. Synthetic preflight tests pass, but neither establishes historical public timing of the 14 October protocol amendment nor recreates the full physical six-arm run. The original six-arm decision prototype remains absent from the supplied archive. The independent_array_check.py script directly reads saved common-grid arrays; it is a newly written retrospective verifier, not the original preregistered execution engine.

3. Endpoint multiplicity and numerical definitions

Twenty-four brackets give 48 *endpoint occurrences*, but some complete parameter settings are reused across different brackets. Therefore these 48 occurrences must not be called 48 independent physical experiments. The supplied saved arrays have 32 distinct rounded (E, tau, common-minimum) numerical fingerprints at seven decimal places. A fingerprint is not by itself proof of a unique physical parameter coordinate; deduplication by law, screening length, beta and all other frozen inputs is required for a definitive count of distinct model evaluations. The previously suggested figure of 33 unique coordinates is therefore a candidate to verify against full parameter metadata, not an unconditional independence claim.

At each common sampled radius, E is the maximum six-arm pointwise spread and tau=max(1e-8, 5E). A robust-positive final outcome requires the minimum F across common- and native-grid minima to exceed tau. A robust-negative outcome additionally requires all six arms to fall below -tau at the same sampled radius. Otherwise the outcome is unresolved; a raw sign change is not a validated strict bracket. E measures inter-configuration numerical spread, not solely round-off error or a proven continuum error bound.

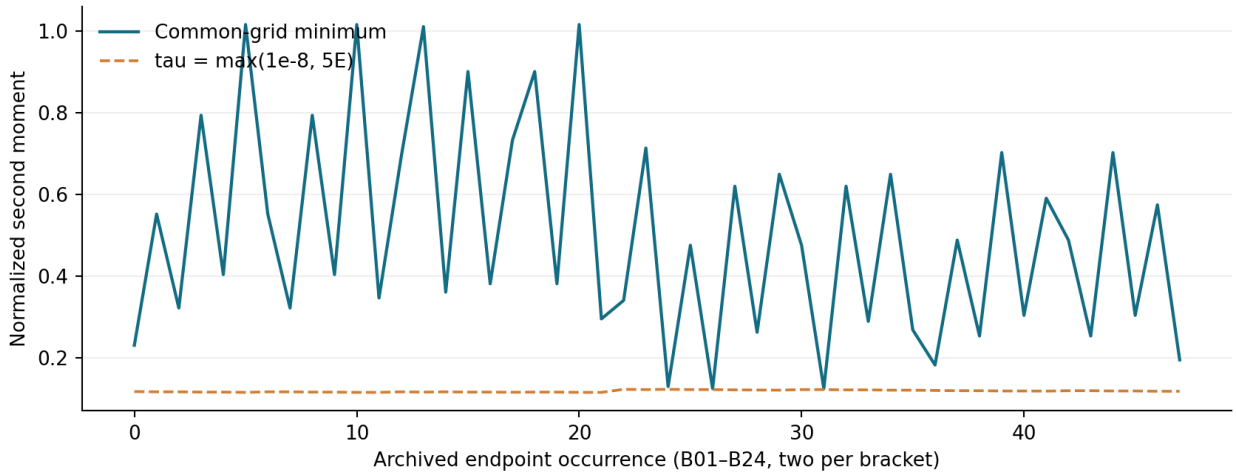


Figure 2. Saved common-domain minima (solid) and tau thresholds (dashed), in original endpoint occurrence order. Native-grid minima are not shown and can change final classifications.

4. Independent array audit and B01 domain mismatch

The independent checker groups 288 saved profiles into 48 six-arm endpoint occurrences, each on 1,200 common sampled radii. It finds no nonfinite profiles, zero six-arm shared-radius negative witnesses and 48 common-grid minima above tau. These results agree with necessary common-domain conditions but do not upgrade the archived 30 unresolved final outcomes.

For B01 at beta=0.35 and screening length 1 pc, the archived native-grid minimum is approximately -0.158964 whereas the saved common-grid minimum is +0.231148, with E=0.0235645 and tau=0.117822. The approximately 0.390 difference is much larger than E, but comparing minima over different radial domains is not a like-for-like discretization error estimate. The exact native and common radial bounds and the native minimum radius must be extracted from original model metadata before attributing the discrepancy to a specific inner radial region. This audit does not establish that the negative native minimum occurs inside or outside a particular numerical radius. The other B01 endpoint also remains unresolved under the

archived full F decision. No B01 strict bracket is validated.

5. Reproducibility evidence and limitations

The post-hoc results support provenance, input integrity and necessary common-domain consistency only. They do not reproduce every native-grid minimum, solve the gravitational field afresh, establish a converged self-consistent 3D total potential Hessian H_ij = partial_i partial_j Phi_total, demonstrate a nonnegative stellar distribution function, or provide direct likelihood validation with Hydrus I individual line-of-sight velocities. No general exclusion of MOND or new gravity law follows.

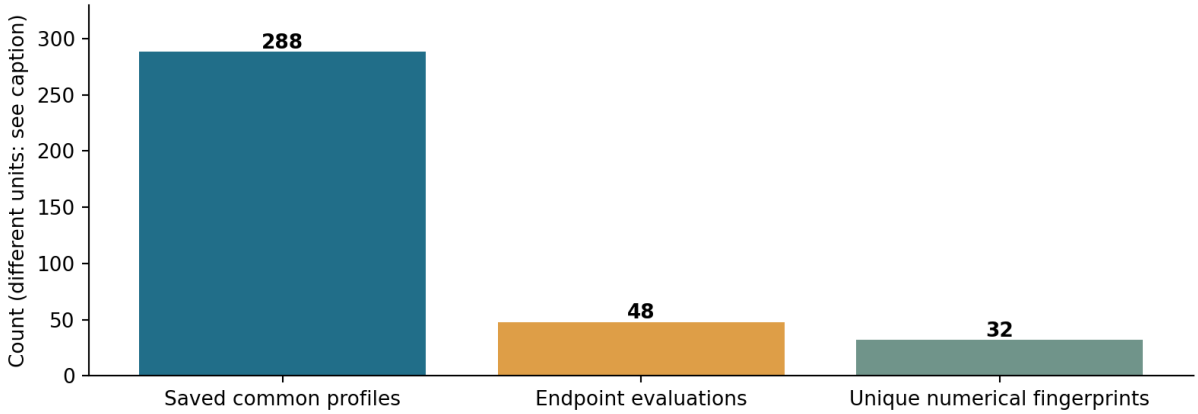


Figure 3. Different counting units: 288 saved arm profiles, 48 endpoint occurrences, and 32 distinct rounded numerical fingerprints. Fingerprint groups are not automatically distinct physical experiments.

6. Proposed v7.14 preregistration preview (not executed)

Before inspecting confirmatory outcomes, v7.14 should publicly freeze (i) the exact Hydrus I member-star catalog, velocity errors, membership probabilities, contamination and binary-star treatment; (ii) explicit likelihood, priors, sampling diagnostics, posterior predictive checks and comparison baselines; (iii) a self-consistent differentiable 3D total-potential solver with boundary conditions and manufactured-solution/convergence controls; (iv) a predeclared Hessian tensor estimator, radial sampling and numerical tolerance; (v) a blinded model comparison and complete negative-result reporting plan. The model-input simulations in v7.10–v7.13 must be distinguished from direct fits to the complete set of individual observed stellar velocities. No v7.14 preregistration or scientific run is claimed here.

7. Conclusion

The central scientific result is bounded but reproducible at the saved-evidence level: the 24 proposed strict sign brackets remain unvalidated, the archived 18/30/0 final endpoint outcomes are not contradicted by an independent common-array check, and zero shared-radius negative witnesses are present in the saved six-arm profiles. The 48 endpoint occurrences are not independent physical tests. The distinction between native- and common-grid minima, unresolved source dependencies and absent 3D Hessian/individual-velocity verification are explicit. The present contribution is an audit of recorded evidence, not a new confirmatory physical result.

References

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Reproducibility and link status

Original Hessian archive: https://github.com/wspaik5/UDCT_v7.13_Postregistration_Hessian_Archive
Original preregistration: https://github.com/wspaik5/UDCT_v7.13_Preregistration_Jeans_Boundary_Refinement
New Post-registration II repository: https://github.com/wspaik5/UDCT_v7.13_Postregistration_II_Reproducibility_Audit
New Post-registration II DOI: https://doi.org/10.5281/zenodo.23559995
See PROVENANCE_AND_AUDIT.json for original archive SHA-256 and provenance.